

WaterForm: Altering the Liquid to Generate Multisensory Feedback for Enhancing Immersive Environment

I-Chin Chen
National Taipei University of
Technology
Taipei, Taiwan
ichinchen1112@gmail.com

Hsin-Chia Chang
National Taipei University of
Technology
Taipei, Taiwan
xin1106.zhang@gmail.com

Chia-Yi Hung
National Taipei University of
Technology
Taipei, Taiwan
chiayiii722@gmail.com

Chi-Yu Lin
National Taipei University of
Technology
Taipei, Taiwan
rich70929@gmail.com

Kuan-Ning Chang
National Taipei University of
Technology
Taipei, Taiwan
kuan19951215@gmail.com

Ping-Hsuan Han
National Taipei University of
Technology
Taipei, Taiwan
pinghsuan.han@gmail.com

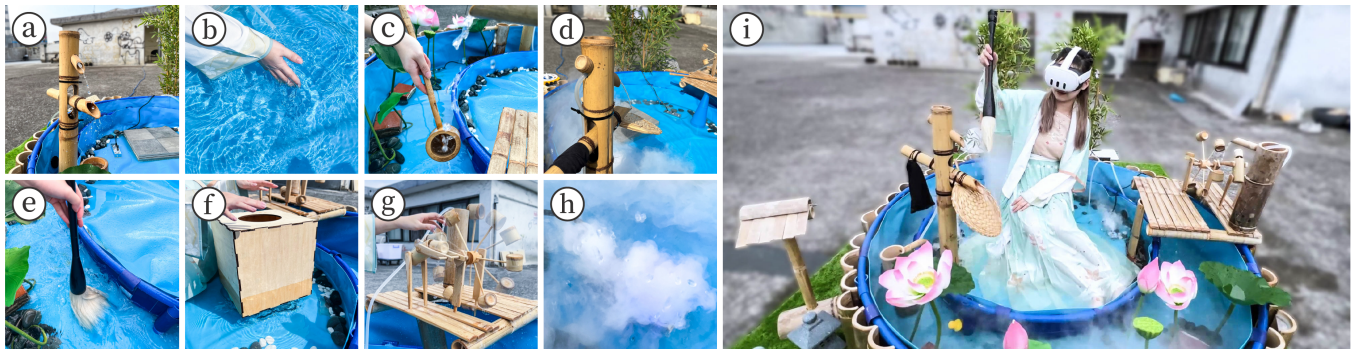


Figure 1: Our liquid-transformation systems produces (a) water splash, (b) water flow, (c) gravity, (d) wind, (e) resistance, (f) buoyant force, (g) mechanical energy, and (h) mist to enhance (i) immersive environment.

ABSTRACT

With virtual reality and haptic technologies, we can create a compelling, immersive experience. Thus, researchers have been exploring integrating different feedback modules to provide more abundantly hybrid feedback systems. Alternatively, they want to use the same module to generate multiple feedback using different interaction techniques. However, simultaneously providing multiple sensory feedbacks in virtual environments presents challenges beyond system integration. Ensuring that users can experience multiple stimuli without compromising the overall immersive experience, as in a physical world, is still challenging. Therefore, we present WaterForm, a liquid-transformation system that utilizes the liquid to generate multisensory feedback, including water splash, water flow, gravity, wind, resistance, buoyant force, mechanical energy, and mist, to enhance the immersive environment. In our

demonstration, we developed a VR excursion in the virtual East landscape to explore self-awareness via immersive storytelling.

CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms**; Virtual Reality.

KEYWORDS

Liquid-transformation Systems, Hybrid Feedback, Relaxation Application, Mixed Reality

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1 INTRODUCTION

The research team has been devoted to providing various tactile experiences in virtual reality (VR) to enhance immersive experiences using different technologies. Additionally, liquids can directly

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or indirectly offer multisensory feedback through shape transformation. GroundFlow [Han et al. 2023] proposes a liquid tactile floor system to simulate fluid on the ground in VR, employing multi-flow feedback mechanisms to allow users to explore rivers, beaches, and swimming pools. Lotus [Chen et al. 2018] utilizes the transformation of liquid into mist, guided by airflow modules to simulate olfactory environments. AoEs [Han et al. 2017] utilize tactile simulation integrated devices to create immersive environments, providing two virtual environments (VE) simultaneously to achieve long-distance transmission experiences. AquaTop Display [Matoba et al. 2013] uses water as the screen surface for projection systems, allowing users to interact immersively both above and below the water surface. In addition to techniques involving direct contact with water, previous research has also explored generating sensory feedback indirectly through the transformation of water. GravityPack [Chen et al. 2022] introduces a wearable gravity display that mimics grasping, holding, and releasing virtual objects in the virtual world using liquid simulation. GravityCup [Cheng et al. 2018] simulates the weight and inertia of real objects using liquid. INVISIBL [Nakano et al. 2006] proposes a weight perception system where users experience weight through fluid transmission. While previous research cases have used various methods to transform water into different types of feedback, there has yet to be a system that provides different feedback simultaneously. One possible approach is to use the same module to generate multiple feedbacks simultaneously through different interaction techniques. Hence, we design and implement a liquid-transformation system, utilizing water as a single module to generate water splash, water flow, gravity, wind, resistance, buoyant force, mechanical energy, and mist to enhance immersive environments via relaxation application to demonstrate the practicality of our device.

2 DESIGN AND IMPLEMENTATION

WaterForm is a liquid-based haptic system that utilizes liquid to generate multisensory feedback, including water splash, water flow, gravity, wind, resistance, buoyant force, mechanical energy, and mist, enhancing the immersive environment (Figure 1i). The total area of our device is 5m*3m, composed of internal and external pools and modules. We store water in the external pool ($\varnothing 1.57\text{m} \times 0.35\text{m}$) to a depth of approximately 0.1m and utilize 4 high-power pumps (BS6500, flow rate: 8.0L/min) to inject water into the internal pool and various modules for transforming the water form. Users sit on diatomaceous earth ($0.4\text{m} \times 0.4\text{m} \times 0.25\text{m}$) stacked inside the internal pool to experience. Water from the Jing-Lu falls from a height, creating a water splash, and the bamboo ladle fills the water to sense gravity (Figure 1a, 1c). The system can generate wind by applying weight behind the Jing-Lu and activating a fan in front of the Jing-Lu through the lever mechanism (Figure 1d). The wheel driven by the water generates mechanical energy (Figure 1g). Waving hands in the pool allows users to feel the water flow (Figure 1b). The Atomizers placed 0.02m underwater generate mist (Figure 1h). Users can experience buoyant force by pressing the box and wielding a brush to sense resistance (Figure 1f, 1e). Our device achieves liquid transformation simulation through Unity and Linkit 7697 via Bluetooth. Additionally, elements of Taoist philosophy are incorporated into the landscape design of the pond. Surrounding the pond are

embellishments such as lotus, bamboo, and stone lanterns, creating an overall serene atmosphere. The pond represents the Tai Chi diagram, with the inner portion representing "yang" and the outer portion representing "yin." The interaction between the two serves as a medium for harmonizing yin and yang, allowing for balance and self-exploration.

3 APPLICATION: RELAXATION APPLICATION

We design the relaxation application with a virtual East landscape to demonstrate our concept. Users will wear the Meta Quest 3 while seated within our device and guided by the voice to experience. Based on our system, we create VE with eight interactive modes in which users can immerse themselves in serene surroundings, enjoying the tranquil scenery, and freely interacting with physical devices from the real world. Guided prompts will direct users to focus on sensory changes in the present moment, facilitating exploration between the virtual and physical realms.

4 DISCUSSION AND FUTURE WORK

Our project introduces a liquid-transformation system to enrich immersive experiences with hybrid tactile feedback, including water splash, flow, gravity, wind, resistance, buoyant force, mechanical energy, and mist. Users wearing the HMD can sit within our device to explore varied haptic in our relaxation application. Moving forward, we will conduct user studies to gain deeper insights into the potential development of flow state, mindfulness, meditation, and other related fields.

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REFERENCES

- Yang-Sheng Chen, Ping-Hsuan Han, Kong-Chang Lee, Chiao-En Hsieh, Jui-Chun Hsiao, Che-Ju Hsu, Kuan-Wen Chen, Chien-Hsing Chou, and Yi-Ping Hung. 2018. Lotus: enhancing the immersive experience in virtual environment with mist-based olfactory display. In *SIGGRAPH Asia 2018 Virtual & Augmented Reality*. 1–2.
- Yu-Yen Chen, Yi-Jie Lu, and Ping-Hsuan Han. 2022. GravityPack: Exploring a Wearable Gravity Display for Immersive Interaction Using Liquid-based System. In *ACM SIGGRAPH 2022 Posters*. 1–2.
- Chih-Hao Cheng, Chia-Chi Chang, Ying-Hsuan Chen, Ying-Li Lin, Jing-Yuan Huang, Ping-Hsuan Han, Ju-Chun Ko, and Lai-Chung Lee. 2018. GravityCup: a liquid-based haptics for simulating dynamic weight in virtual reality. In *Proceedings of the 24th ACM Symposium on Virtual Reality Software and Technology*. 1–2.
- Ping-Hsuan Han, Chiao-En Hsieh, Yang-Sheng Chen, Jui-Chun Hsiao, Kong-Chang Lee, Sheng-Fu Ko, Kuan-Wen Chen, Chien-Hsing Chou, and Yi-Ping Hung. 2017. AoEs: Enhancing teleportation experience in immersive environment with mid-air haptics. In *ACM SIGGRAPH 2017 Emerging Technologies*. 1–2.
- Ping-Hsuan Han, Tzu-Hua Wang, and Chien-Hsing Chou. 2023. GroundFlow: Liquid-based Haptics for Simulating Fluid on the Ground in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics* 29, 5 (2023), 2670–2679.
- Yasushi Matoba, Yoichi Takahashi, Taro Tokui, Shin Phuong, Shingo Yamano, and Hideki Koike. 2013. AquaTop display: a true" immersive" water display system. In *ACM SIGGRAPH 2013 Emerging Technologies*. 1–1.
- Yoshinobu Nakano, Haruka Ouwa, Kenji Kawashima, Hitomi Komura, Masayuki Sagano, Taku Shirazawa, Soshi Tsujimura, Masako Fukushima, Masayuki Yamazaki, Yasuhiro Yoshimura, et al. 2006. INVISIBLE: the shadow chaser. In *ACM SIGGRAPH 2006 Emerging technologies*. 19–es.